

Imformer:

an implicit model of attention

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Abstract

An attention mechanism is integrated with embeddings jointly learned from an implicit deep learning model.

ImFormer - an implicit model of attention

For token string $\{t_1, \dots, t_m\} \subset \mathbb{R}^r$ learn matrices

K, Q, V, Q' and fixed points $s_i \in \mathbb{R}^n$, $n > r$,

a latent token, we call a Noken, so that:

$$s_i = \varphi(Ks_i + Qt_i)$$

$$\varepsilon = Vs_i + Q't_i$$

where hyperparameter $\varepsilon \in \mathbb{R}^n$ and

$$K = \theta - \text{diag}(|\theta| \mathbb{1}_n) + \delta \mathbb{I}_n, \quad 0 < \delta < 1,$$

$$\theta \in \mathbb{R}^{n \times n}, \quad \mathbb{1}_n = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1} \quad \text{and} \quad \mathbb{I}_n \in \mathbb{R}^{n \times n} \text{ the identity matrix,}$$

ensures K is well-posed for φ , that is

fixed points s_i exist and are unique [2].

Let $k_i = Ks_i$, $q_i = Qt_i$, $v_i = Vs_i$ and $q'_i = Q't_i$.

Let score be a similarity function eg a kernel.

Define probabilities $\alpha_{ij} = \text{softmax}_{j \in [m]} \text{score}(q'_i, k_j)$, $i \in [m]$

and average Renyi α -entropy

$$E_{\varepsilon} = \frac{1}{m} \sum_{i=1}^m H_{\alpha}^i, \quad H_{\alpha}^i = H_{\alpha}(\{\alpha_{ij}\}_{j=1}^m) \text{ and}$$

H_{α} is Renyi α -entropy.

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Noken Rademacher descent [3]

Hyperparameters: step size $\eta > 0$

of iterations N

Initial step: Set $S = 0 \in \mathbb{R}^n$

Define $\varepsilon_i = (\sigma_i, \dots, \sigma_n)$ where σ_i are independent draws

$$\text{Prob}(\sigma_i = S + \eta) = \frac{1}{2} = \text{Prob}(\sigma_i = S - \eta)$$

Induction step: Given $\mathcal{E}_k$, set $\mathcal{S} = \mathcal{E}_k$ and

draw $\mathcal{E}' = (\sigma'_1, \dots, \sigma'_n)$ where σ'_i are independent

$$\text{Prob}(\sigma'_i = \mathcal{S} + \eta) = \frac{1}{2} = \text{Prob}(\sigma'_i = \mathcal{S} - \eta).$$

$$\text{Let } \mathcal{E}_{k+1} = \begin{cases} \mathcal{E}' & \text{if } E_{\mathcal{E}'} < E_{\mathcal{E}_k} \\ \mathcal{E}_k & \text{otherwise} \end{cases}$$

Iterate N times, to learn implicit model with $\mathcal{E}_N$
that potentially has lower average Renyi α -entropy:

$$s_i = \varphi(Ks_i + Qt_i)$$

$$\mathcal{E}_N = Vs_i + Q't_i$$

$$\text{Let } k_i = Ks_i, q_i = Qt_i, v_i = Vs_i \text{ and } q'_i = Q't_i.$$

Define probabilities $\alpha_{ij} = \text{softmax}_{j \in [m]} \text{score}(q'_i, k_j)$, $i \in [m]$

and average Renyi α -entropy $E_{\mathcal{E}_N}$, as before.

Attention updating

Define $h_i = \text{FFN}_\theta \left(\sum_{j=1}^m \alpha_{ij} v_j \right)$, $i \in [m]$

where FFN_θ is a feed-forward network with parameters θ

Define probabilities $\alpha'_{ij} = \underset{j \in [m]}{\text{softmax}} \text{score}(h_i, k_j)$, $i \in [m]$

and $j_i^{\alpha'} = \arg \max_{j \in [m]} \alpha'_{ij}$.

Given input token t_i , define output $o_{i+1} = t_{j_i^{\alpha'}}$.

Learn θ to minimize perplexity $(o_2, \dots, o_{m+1})$.

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Define update $h'_i \triangleq \text{FFN}_{\theta'} \left(\sum_{j=1}^m \alpha'_{ij} v_j \right)$

and let $\alpha''_{ij} \triangleq \underset{j \in [m]}{\text{softmax}} \text{score}(h'_i, k_j)$, $i \in [m]$

and $j_i^{\alpha''} \triangleq \arg \max_{j \in [m]} \alpha''_{ij}$, $o'_{i+1} \triangleq t_{j_i^{\alpha''}}$, as before.

Learn θ' to minimize perplexity $(o'_2, \dots, o'_{m+1})$

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Repeat updating until perplexity is acceptable,

as with transformer attention.

Summary

Given token string $t_1, \dots, t_m$ an implicit model [1]

$$s_i = \varphi(Ks_i + Qt_i)$$

$$\varepsilon = Vs_i + Q't_i$$

jointly learns embeddings K, Q, V, Q'

such that $\{q'_i\}_{i=1}^m$ has reduced average Renyi

α -entropy after Noken Radermacher descent on ε .

Then the contextual representations h_i , of token t_i ,

are updated by feed-forward networks to reduce the

perplexity of $\{o_{i+1}\}_{i=1}^m$, where o_{i+1} is the token

whose key is most similar to h_i . This is attention

achieved on embeddings jointly learned from an

implicit deep learning model.

□

References

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